MIL-B-0051097D(EA)
8 March 1976
SUPERSEDING
MIL-B-0051097C(MU)
22 July 1968
USED IN LIEU OF
MIL-B-51097B
30 September 1964

MILITARY SPECIFICATION

BACKING MATERIAL, C18

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-B-51097B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers C18 and C18R1 backing material.

1.2 Classification. The backing material shall be furnished as specified (see 6.2).

C18 - 6 inch minimum water repellency.
C18R1 - 18 inch minimum water repellency.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-P-268 - Paper, Kraft, Wrapping.
MMM-A-250 - Adhesive, Water-Resistant (for Closure of Fiberboard Boxes).
MMM-A-260 - Adhesive, Water-Resistant, (For Sealing Water-proofed Paper).

FSC 4240

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PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291 - Paperboard, Wrapping and Cushioning.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-F-51095 - Filter Material, Gas-Aerosol, C18.

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

DLB125-8-1 - Q101 Water Repellency Indicator.
DLB136-42-102-5B - Q127 DOP Filter Testing Penetrometer.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D645 - Thickness of Paper and Paperboard.
D646 - Basis Weight of Paper and Paperboard.
D828 - Tensile Breaking Strength of Paper and Paperboard.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. The backing material shall be manufactured from the following fibers (see 6.3):

(a) Glass: 0.50 to 0.75 micrometer diameter.
(b) Vinyl chloride-vinyl acetate copolymer: 85 to 88 percent vinyl chloride, 12 to 15 percent vinyl acetate; 3 denier.
(c) Regenerated cellulose: 1.5 denier.

3.2 Physical properties. The physical properties of the backing material shall conform to table I when tested as specified therein.

3.3 Size. The backing material shall be furnished in rolls with a maximum of three splices permitted for each roll. Each splice shall be lapped (approximately 1.5 inches) and heat sealed across the width. The location of the splices shall be indicated by a paper "flag" of contrasting color. The width and diameter (including tolerance) of the roll shall be as specified at the time of procurement (see 6.2).

3.4 Workmanship. The backing material shall be free from foreign matter, tears, punctures, pickouts, pull aparts, holes, thick or thin spots, burn holes, charring, scorching, wrinkles, pills, or transparencies.

3.5 Mildew inhibitor. The backing material shall contain 1.0 $\pm$ 0.1 percent mildew inhibitor based on the weight of viscose rayon present when applying mildew inhibitors "Zerlate Ziram" (supplied by DuPont), "Berchem Ziram WP" (supplied by Bercen Chemical Co., Providence, RI), or "Methyl Ziram 76%" (supplied by Fike Chemical of Nitro, West Virginia). When "Sunsize 226B" is used as the water-repellent, 3.0 $\pm$ 0.5 percent "GAF Preventol GDC" mildew inhibitor based upon fiber weight shall be put into the slurry solution.

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Table I. Physical properties

Property	Requirement	Test paragraph
Air resistance, mm water	*	4.2.4.1
DOP penetration, %	*	4.2.4.1
Thickness, inches	0.020 to 0.030	4.2.4.2
**Water repellency, inches	*	4.2.4.3
Average tensile strength, lbs/in		4.2.4.4
machine direction	1.5 (min)	
cross direction	1.0 (min)	
***Basis weight, lbs/3000 sq. ft.	70 to 80	4.2.4.5
Moisture content, %	5.0 (max)	4.2.4.6

*The Government or contractor, whichever is applicable, shall specify the air flow resistance and DOP smoke penetration (see 6.2 and 6.7) and water repellency (see 6.4) at the time of procurement in order to meet the requirements of MIL-F-51095.

**Responsibility for treatment for water repellency and approval of method shall be specified in the contract or order (see 6.2).

***The basis weight of two consecutive samples, taken five minutes apart without adjustment to the equipment, shall be determined at the start of production. The material on the reel shall be flagged to indicate the start of acceptable material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.3, 3.5 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the backing material produced during one work shift not to exceed 12 hours, by one manufacturer, from the same materials, and under essentially the same manufacturing conditions.

4.2.2 Sampling.

4.2.2.1 For examination. A six-foot long sample, full width, shall be taken from each roll. Each sample shall be identified with the roll represented.

4.2.2.2 For tests. An appropriate size strip shall be taken across the width of each sample (4.2.2.1). Each strip shall be divided into specimens which shall be identified with the roll represented.

4.2.3 Examination.

4.2.3.1 Rolls. Prior to reeling, a visual examination of the backing material shall be conducted over a source of light under the full width of the reel. Ten square yards representing each component roll shall be examined for defects in accordance with table II. Re-examination shall be made on an additional 10 square yards if permitted by table II. In addition, each 6-foot sample shall be examined over a light box in accordance with table II, except that re-examination shall not be allowed.

4.2.3.2 Preparation for delivery. Sample reels shall be examined for the level B preparation for delivery requirements in accordance with the classification of defects and with MIL-STD-105.

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Table II. Examination

Defects	Number of defects: 1st sample	Number of defects: 1st and 2nd sample combined	Action per roll
Tears, punctures, holes, pick outs or pull-aparts (exceeding 1/16 inch)	1	2	Accept
	2		Re-examine
	3 or more	3	Reject
Thick or thin spots	1	2	Accept
	2		Re-examine
	3 or more	3	Reject
Burn holes, charring or scorching	0 (max)		Accept
Wrinkles	0 (max)		Accept
Pills or fiber agglomerates greater than 3/16 inch	10/square foot (max)		Accept
Transparencies (maximum dimension at least 1/4 inch)	1/square foot (max)		Accept
Foreign matter, exceeding an area of 1/4 inch by 1/4 inch	1	2	Accept
	2		Re-examine
	3 or more	3	Reject

4.2.3.3 Classification of defects.

(a) Preparation for delivery.

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Spirally wound core not as specified
102	Initial, intermediate or final wrap not as specified.
103	Circular pack not as specified.
104	Circular pads missing or incorrectly placed.
105	Sealing material missing or inadequate.
106	Marking, missing, incorrect or illegible.

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4.2.4 Tests. Each roll sample taken as specified in 4.2.2.2 shall be tested in accordance with table III. Test specimens shall be taken at random from each sample.

Table III. Tests

Test	Test paragraph	Test specimens at each end of roll*	Defective specimens at same end	Action per roll	Retest specimen at end that failed	Total defective specimen	Action per roll
Airflow resistance and DOP smoke penetration.	4.2.4.1	3	0	Accept	3	1	Accept
			1	Retest			
			2	Reject		2	Reject
Thickness	4.2.4.2	3	0	Accept	3	1	Accept
			1	Retest			
			2	Reject		2	Reject
Water repellency	4.2.4.3	3	0	Accept	3	1	Accept
			1	Retest			
			2	Reject		2	Reject
Tensile strength	4.2.4.4	3 in each direction	Reject the roll if the average in either direction does not meet the tensile strength requirement.				
Basis weight	4.2.4.5	2	0	Accept	2	1	Accept
			1	Retest			
			2	Reject		2	Reject
Moisture content	4.2.4.6	1	0	Accept	2	1	Accept
			1	Retest			
						2	Reject

*If the end roll is adjacent to the beginning of the following roll, specimens need be taken only at one end and the results used for both rolls.

4.2.4.1 Air flow resistance and DOP smoke penetration. The test specimens (5.25-inch-diameter disks) shall be tested for airflow resistance and DOP smoke penetration at a flow rate of 32 liters per minute, with Q127 DOP Filter Testing Penetrometer.

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4.2.4.2 Thickness. Thickness of the test specimens shall be determined in accordance with ASTM D645, except that conditioning of specimens shall be omitted. Three determinations shall be made on each specimen and the average reported as the thickness.

4.2.4.3 Water repellency. The test specimens (2.75- by 5.50-inch rectangles) shall be tested for water repellency with Q101 Water Repellency indicator. The two surfaces of each test specimen shall be identified as top and bottom. The specimen shall then be cut into two equal squares. The top surface of one square and the bottom surface of the other square shall be tested. The lesser of the two results shall be considered the water repellency of the specimen.

4.2.4.4 Tensile strength. Tensile strength of the test specimens shall be determined in accordance with ASTM D828 except that conditioning of specimens shall be omitted.

4.2.4.5 Basis weight. Basis weight shall be determined in accordance with ASTM D646, except the conditioning of specimens shall be omitted.

4.2.4.6 Moisture content. Each test specimen (5.25-inch-diameter disk) shall be weighed to the nearest 0.01 gram (wet weight). The specimen shall be placed in a forced air drying oven at 105° to 110° C until constant dry weight is reached. The final weighing (dry weight) shall be recorded to the nearest 0.01 gram. The percent moisture shall be calculated as follows:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{Dry weight})}{\text{wet weight}}$$

5. PREPARATION FOR DELIVERY

5.1 Packing. Levels shall be as specified (see 6.2).

5.1.1 Level B. The backing material shall be evenly wound onto a spirally wound hollow core with a diameter of 3 ± 1/8 inch and a minimum 3/8 inch wall thickness. The hollow core shall be constructed of fiberboard or other suitable material. The individual rolls of core material shall be initially wrapped with type I, grade B, 40-pound basis weight paper conforming to UU-P-268; the paper shall be secured in place with minimum 2 inch wide tape conforming to PPP-T-76. Each roll shall then be intermediately wrapped with a minimum of three complete wraps of type III, style 2, paperboard conforming to PPP-P-291; the paperboard shall be secured in place with the aforementioned tape. Four circular pads, fabricated from type CF, class weather resistant, variety SW, grade

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V3c fiberboard conforming to PPP-F-320, shall be laminated together with adhesive conforming to MMM-A-250 and placed at each end of the wrapped roll. The circular fiberboard pads shall have the same diameter as the overall diameter of the intermediately wrapped rolls; each circular fiberboard pad shall be secured in place with a minimum of ten strips of 2 inch wide by 12 inch long tape conforming to type IV of PPP-T-97. The ten strips of tape shall be equally spaced around the circumference of the fiberboard pad. Each roll shall then be overwrapped with class E-1 barrier material conforming to PPP-B-1055; and, all folds, laps, and seams in the paper wrapping shall be sealed in place with adhesive conforming to MMM-A-260.

5.1.2 Commercial packing. The material shall be packed in accordance with FED-STD-356.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Commercial shipping containers shall be marked in accordance with FED-STD-356.

5.2.1 Special marking. The following special marking shall be minimum 1 inch high letters, and shall be applied in two places on the body of the roll in such location so as not to interfere with other specified marking:

"FRAGILE, HANDLE WITH CARE,

DO NOT DROP"

6. NOTES

6.1 Intended use. This specification covers backing material intended to be used as a component of either the C18 or C18R1 Gas-Aerosol Filter Material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Roll width and diameter including tolerances.
(c) Responsibility for treatment for water repellency and for approval of water repellency method.
(d) Level of packing required.

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(e) Percent DOP penetration and air-flow resistance (see 6.7) and water repellency (see 6.4).
(f) Whether C18 or C18R1 backing material is required.

6.3 Composition. The following materials have been found to be satisfactory:

AAA 475-106 J-M glass fibers	- 15 percent by weight
Vinyon HH (1/8 or 1/4-inch staple, 3 denier)	- 42.5 percent by weight
Viscose rayon (1/4-inch staple max, 1.5 denier)	- 42.5 percent by weight

6.4 Water repellency. The C18 backing material shall have a water repellency of 8 inches minimum. The C18R1 backing material shall have a water repellency of 18 inches minimum and 20 inch lot average. Any satisfactory method for the application of water repellent compound to the backing material may be used after approval by the procuring agency. An approved method using methyl trichlorsilane is described in Edgewood Arsenal Directive 233; an alternate material, "Sunsize 226B", has been found to provide adequate water repellency.

6.5 For non-military procurement. Approval shall be obtained from the procuring agency.

6.6 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.7 Core and backing material. Core and backing materials having the following physical properties are recommended for the manufacture of type II or III C18 or C18R1 Gas Aerosol Filter Material covered by MIL-F-51095.

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Percent of each shipment	Core material: For type II or III: Percent DOP	Core material: For type II or III: Air resistance	Backing material: For type II: Percent DOP	Backing material: For type II: Air resistance	Backing material: For type III: Percent DOP	Backing material: For type III: Air resistance
85 min	70 max	2.7 max	5 max	13 max	8 max	12 max
15 max	80 max	3.0 max	8 max	14 max	10 max	13 max

Custodian:

Army - EA

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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